

INTEREST

Banks pay interest on the money that savers invest with them (capital), and charge interest on money that is borrowed from them. Interest is given as a percentage, and there are two types, simple and compound.

Simple interest

This is interest paid only on the sum of money that is first saved with the bank. If \$10,000 is put in a bank account with an interest rate of 0.03, the amount will increase by the same figure each year.

amount saved (capital) interest rate number of years

$$\text{Interest} = P \times R \times T$$

△ Simple interest formula

To find the simple interest made in a given year, substitute real values into this formula.

First year

amount invested interest rate number of years interest

$$10,000 \times 0.03 \times 1 = 300$$

Substitute the values in the formula to work out the value of the interest for the year.

invested amount interest total

$$10,000 + 300 = 10,300$$

After one year, this is the total amount of money in the saver's bank account.

Second year

result is the same

$$10,000 \times 0.03 \times 1 = 300$$

Substitute the values in the formula to work out the value of the interest for the year.

starting amount interest total

$$10,300 + 300 = 10,600$$

After two years the interest is the same as the first year, as it is only paid on the initial investment.

Compound interest

This is where interest is paid on the money invested and any interest that is earned on that money. If \$10,000 is paid into a bank account with an interest rate of 0.03, then the amount will increase as follows.

amount after T years amount saved interest rate number of years

$$\text{Amount} = P(1 + R)^T$$

△ Compound interest formula

To find the compound interest made in a given year, substitute values into this formula.

First year

amount invested interest rate number of years total

$$10,000 \times (1 + 0.03)^1 = 10,300$$

Substitute the values in the formula to work out the total for the first year.

total after first year original amount interest

$$10,300 - 10,000 = 300$$

After one year the total interest earned is the same as that earned with simple interest (see above).

Second year

number of years

$$10,000 \times (1 + 0.03)^2 = 10,609$$

Substitute the values in the formula to work out the total for the second year.

total after second year total after first year interest

$$10,609 - 10,300 = 309$$

After two years there is a greater increase because interest is also earned on previous interest.